

Wave 3 Statistical Analyses

Marta Topor & Bertil Wegmann

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Hierarchical Bayesian Models Wave 3

This document runs the models used to analyse the main outcomes for Wave 3.
The structure of the document follows the Analysis_Plans.txt file.
The variables in the datasets are explained in Meta-Data.txt.

Data Set 1 - Binary Categorical Outcomes

Crisis aware

Bayesian Logistic regression for whether participants were aware of the reproducibility crisis or not (yes/no).
The explainer variables are:

* Whether the participant was in their first, second, placement or third year

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##           mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu          0.20    0.02 0.93   -1.65   -0.39    0.19    0.79    2.08 2559
## sigma        0.68    0.00 0.31    0.27    0.46    0.61    0.82    1.46 4771
## alpha[1]    -0.39    0.02 1.02   -2.43   -1.04   -0.39    0.26    1.64 2684
## alpha[2]     0.29    0.02 1.01   -1.68   -0.36    0.27    0.92    2.34 2604
## alpha[3]     0.94    0.02 1.02   -1.03    0.28    0.91    1.58    3.04 2617
## alpha[4]    -0.04    0.02 1.01   -2.02   -0.69   -0.05    0.59    2.00 2625
## alpha[5]     0.49    0.02 1.03   -1.43   -0.19    0.45    1.15    2.64 2952
## alpha[6]     0.19    0.01 0.66   -1.08   -0.25    0.19    0.63    1.51 3088
## alpha[7]    -0.09    0.02 1.00   -2.12   -0.73   -0.08    0.55    1.85 2969
## beta1       -0.58    0.02 1.04   -2.71   -1.25   -0.57    0.10    1.44 2636
## beta2       -0.20    0.01 0.50   -1.21   -0.52   -0.18    0.14    0.76 2824
## beta3        0.84    0.01 0.54   -0.19    0.48    0.82    1.19    1.94 2983
## lp__       -333.85    0.05 2.89  -340.45  -335.55  -333.46  -331.74  -329.31 4042
##           Rhat
## mu             1
## sigma          1
## alpha[1]       1
## alpha[2]       1
## alpha[3]       1
## alpha[4]       1
## alpha[5]       1
## alpha[6]       1
## alpha[7]       1
```

```
## beta1      1
## beta2      1
## beta3      1
## lp__       1
##
## Samples were drawn using NUTS(diag_e) at Thu Sep 26 08:37:51 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 525

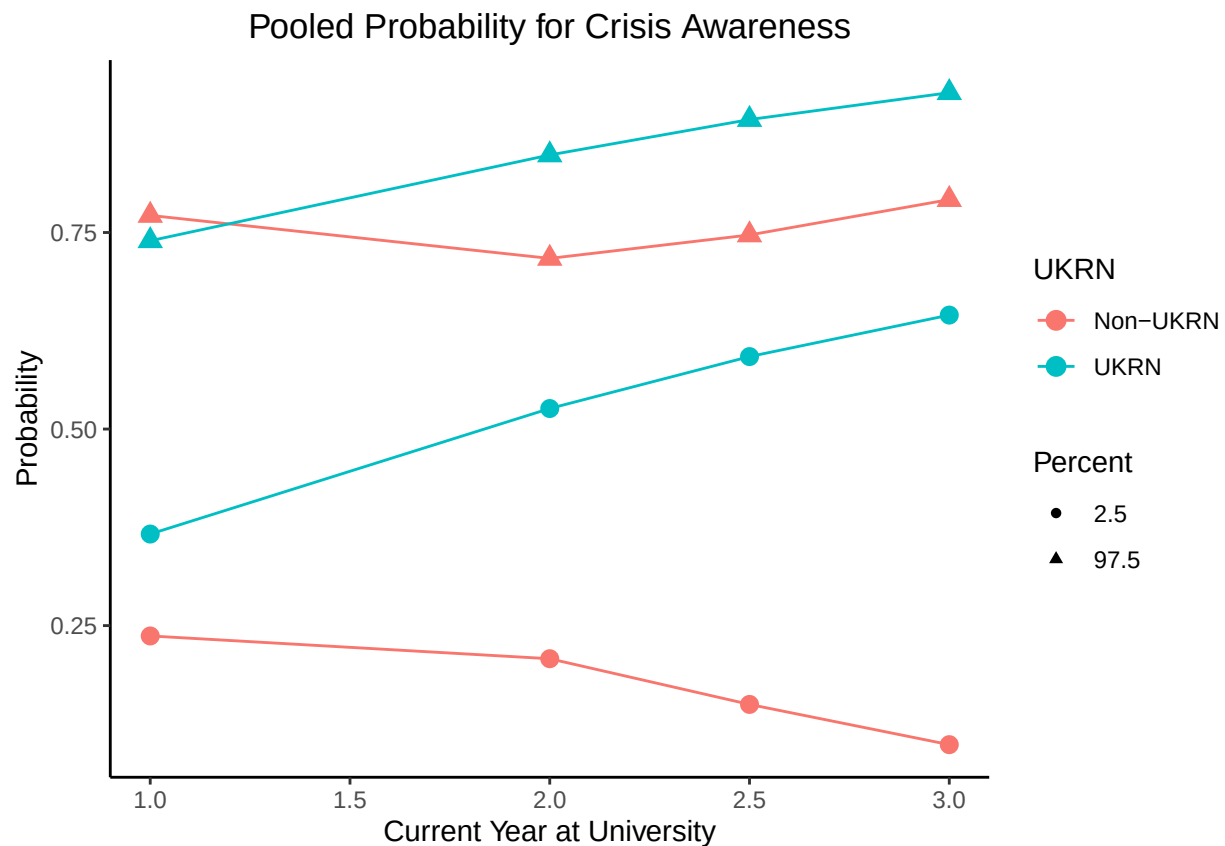
Year 1 N = 316

Year 2 N = 192

Placement Year N = 0

Year 4 N = 17

The output shows no significant results.



Learned about Open Science at Uni

Bayesian Logistic regression for whether participants received teaching about open science or not (yes/no).

The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement or third year * Interaction between the two variables above

```

## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##           mean se_mean   sd   2.5%   25%   50%   75%   97.5% n_eff
## mu          88.90    11.98 53.68    0.63   40.38 102.47 121.52 194.82    20
## sigma        0.26     0.08  0.26    0.04    0.07  0.20   0.35   0.94    12
## alpha[1]     88.79    12.03 53.71    0.50   40.33 102.09 121.51 194.80    20
## alpha[2]     88.94    11.97 53.68    0.69   40.47 102.41 121.55 194.89    20
## alpha[3]     89.08    11.91 53.65    0.85   40.64 102.62 121.58 195.07    20
## alpha[4]     88.80    12.02 53.70    0.53   40.37 102.19 121.51 194.86    20
## alpha[5]     88.88    12.00 53.69    0.44   40.38 102.43 121.54 194.89    20
## alpha[6]     88.96    11.96 53.67    0.73   40.42 102.53 121.52 195.05    20
## alpha[7]     88.88    11.99 53.69    0.52   40.37 102.47 121.55 194.84    20
## beta1       -89.53    11.98 53.68 -195.71 -122.22 -103.06 -41.06  -1.23    20
## beta2       -92.11    11.94 53.66 -197.99 -124.72 -105.69 -43.76  -3.85    20
## beta3        92.50    11.94 53.66   4.16   44.17 106.16 125.12 198.37    20
## lp__       -327.15     2.70  5.51 -337.74 -330.97 -327.64 -322.84 -317.47     4
##           Rhat
## mu           1.23
## sigma        1.14
## alpha[1]     1.23
## alpha[2]     1.23
## alpha[3]     1.23
## alpha[4]     1.23
## alpha[5]     1.23
## alpha[6]     1.23
## alpha[7]     1.23
## beta1        1.23
## beta2        1.23
## beta3        1.23
## lp__         1.41
##
## Samples were drawn using NUTS(diag_e) at Thu Sep 26 09:18:39 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 525

UKRN N = 473

Non-UKRN N = 52

Year 1 N = 316

Year 2 N = 192

Placement Year N = 0

Year 4 N = 17

There were no errors here but the models took a long time to process and the result shows high Rhat values. The results also look quite strange with very large values for Beta 1 and Beta 2. There are conflicting findings in the output

The output shows that beta credibility intervals do not cross 0 for Beta 1 but in the negative direction. This seems to suggest that likelihood of learning about open science is larger in universities that are part of the DKRN.

2.5% = -197.994558

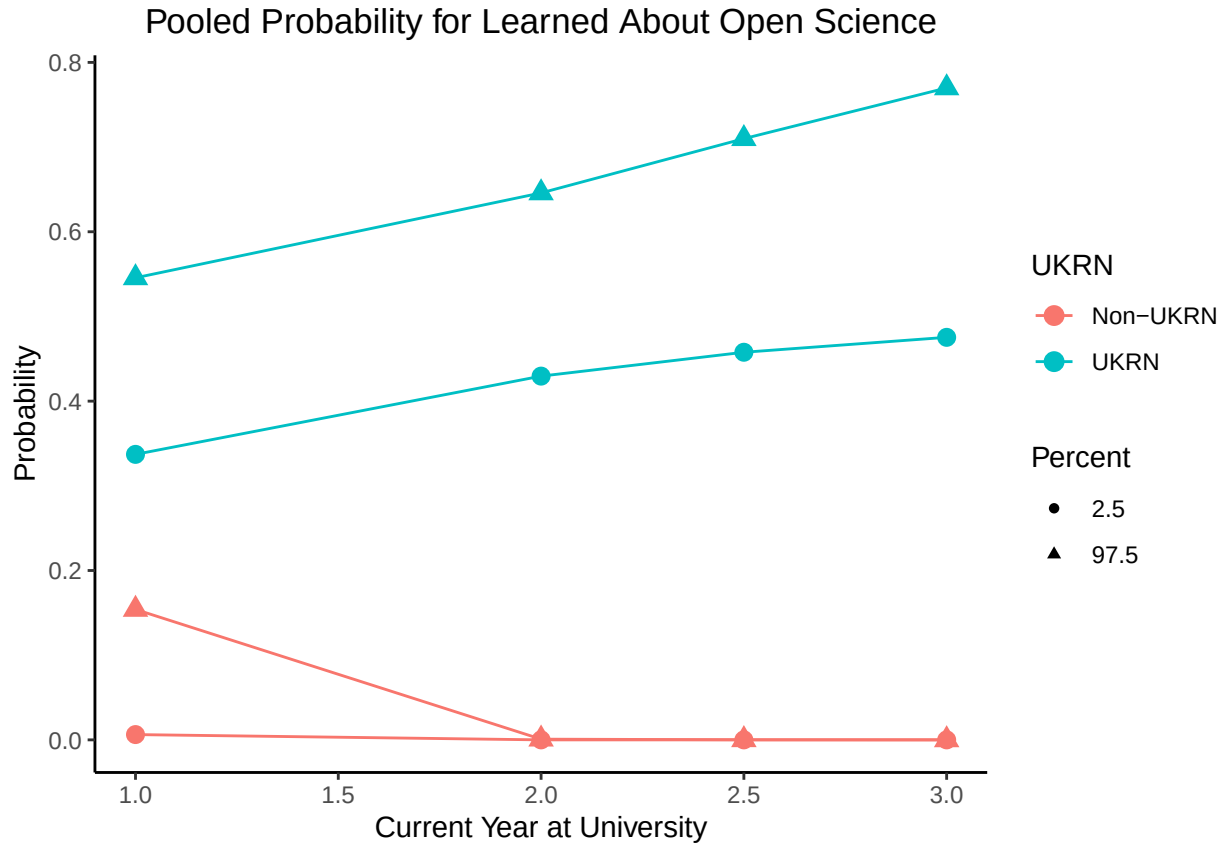
97.5% = -3.8519935

The output shows that beta credibility intervals do not cross 0 for Beta 2 but in the negative direction. This seems to suggest that likelihood of learning about open science is larger in earlier years of education at university

2.5% = -197.994558

97.5% = -3.8519935

There is also a significant interaction for Beta 3 in the positive direction suggesting that the likelihood of learning about open science goes up with the year of study for UKRN universities. 2.5% = 4.1643714
97.5% = 198.3672535



Can explain open science

Bayesian Logistic regression for whether participants can explain what open science is (yes/no).

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement of third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
```

```
## 4 chains, each with iter=10000; warmup=5000; thin=1;
```

```
## post-warmup draws per chain=5000, total post-warmup draws=20000.
```

```
##
```

##	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff
## mu	-0.62	0.05	1.13	-2.63	-1.35	-0.74	0.06	1.82	471
## sigma	0.29	0.02	0.28	0.04	0.10	0.20	0.37	1.03	190
## alpha[1]	-0.78	0.05	1.15	-2.88	-1.48	-0.90	-0.08	1.69	556

```

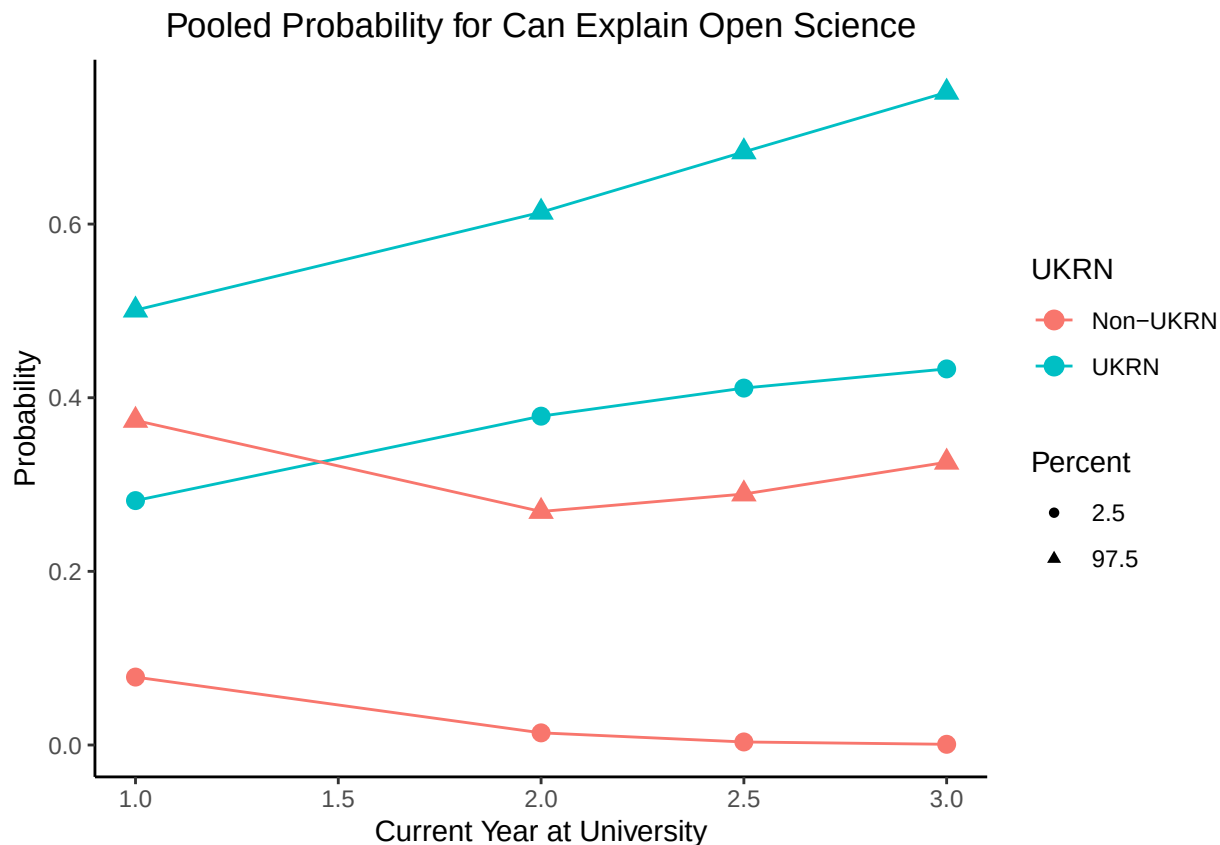
## alpha[2]    -0.56    0.05 1.15   -2.59   -1.32   -0.68    0.15    1.90   439
## alpha[3]    -0.51    0.06 1.15   -2.53   -1.30   -0.64    0.21    1.96   408
## alpha[4]    -0.63    0.05 1.15   -2.67   -1.36   -0.74    0.07    1.85   487
## alpha[5]    -0.47    0.06 1.19   -2.52   -1.29   -0.60    0.27    2.16   394
## alpha[6]    -0.62    0.05 1.05   -2.43   -1.32   -0.73    0.02    1.70   377
## alpha[7]    -0.78    0.05 1.19   -3.03   -1.48   -0.89   -0.07    1.71   610
## beta1       -0.22    0.08 1.20   -2.85   -0.96   -0.12    0.65    1.86   235
## beta2       -0.83    0.04 0.84   -2.93   -1.28   -0.67   -0.27    0.49   390
## beta3        1.24    0.06 0.88   -0.11    0.61    1.11    1.73    3.45   241
## lp__        -340.90   0.85 5.43 -351.35 -344.66 -341.06 -337.26 -330.14   41
##              Rhat
## mu           1.01
## sigma        1.02
## alpha[1]     1.01
## alpha[2]     1.02
## alpha[3]     1.02
## alpha[4]     1.01
## alpha[5]     1.02
## alpha[6]     1.02
## alpha[7]     1.01
## beta1        1.02
## beta2        1.02
## beta3        1.02
## lp__         1.07
##
## Samples were drawn using NUTS(diag_e) at Thu Sep 26 09:23:21 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 525
 UKRN N = 473
 Non-UKRN N = 52
 Year 1 N = 316
 Year 2 N = 192
 Placement Year N = 0
 Year 4 N = 17

The output shows that all beta credibility intervals do cross 0. No significant results.



Applicability

Bayesian Logistic regression for whether participants think that the issues discussed in the questionnaire are relevant to them (yes/no). The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement of third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%    97.5% n_eff
## mu           1.37    0.03 0.89   -0.41    0.79    1.39    1.96    3.11  1213
## sigma        0.33    0.01 0.24    0.06    0.16    0.27    0.43    0.94   802
## alpha[1]     1.11    0.03 0.96   -0.92    0.51    1.13    1.78    2.90  1042
## alpha[2]     1.51    0.02 0.92   -0.34    0.93    1.53    2.08    3.31  1426
## alpha[3]     1.44    0.02 0.92   -0.43    0.86    1.46    2.01    3.24  1384
## alpha[4]     1.44    0.02 0.92   -0.43    0.86    1.45    2.01    3.23  1354
## alpha[5]     1.30    0.03 0.95   -0.64    0.69    1.32    1.94    3.09  1235
## alpha[6]     1.47    0.02 0.78   -0.09    0.95    1.47    1.99    3.01  1245
## alpha[7]     1.34    0.03 0.93   -0.56    0.74    1.35    1.96    3.15  1296
## beta1        -0.80    0.02 0.97   -2.71   -1.39   -0.81   -0.19    1.14  1733
## beta2         0.00    0.01 0.52   -0.95   -0.36   -0.03    0.32    1.10  1336
## beta3         0.65    0.01 0.57   -0.54    0.29    0.67    1.02    1.74  2078
## lp__        -244.20    0.24 4.42  -253.12 -247.11 -244.24 -241.21 -235.55   328
```

```

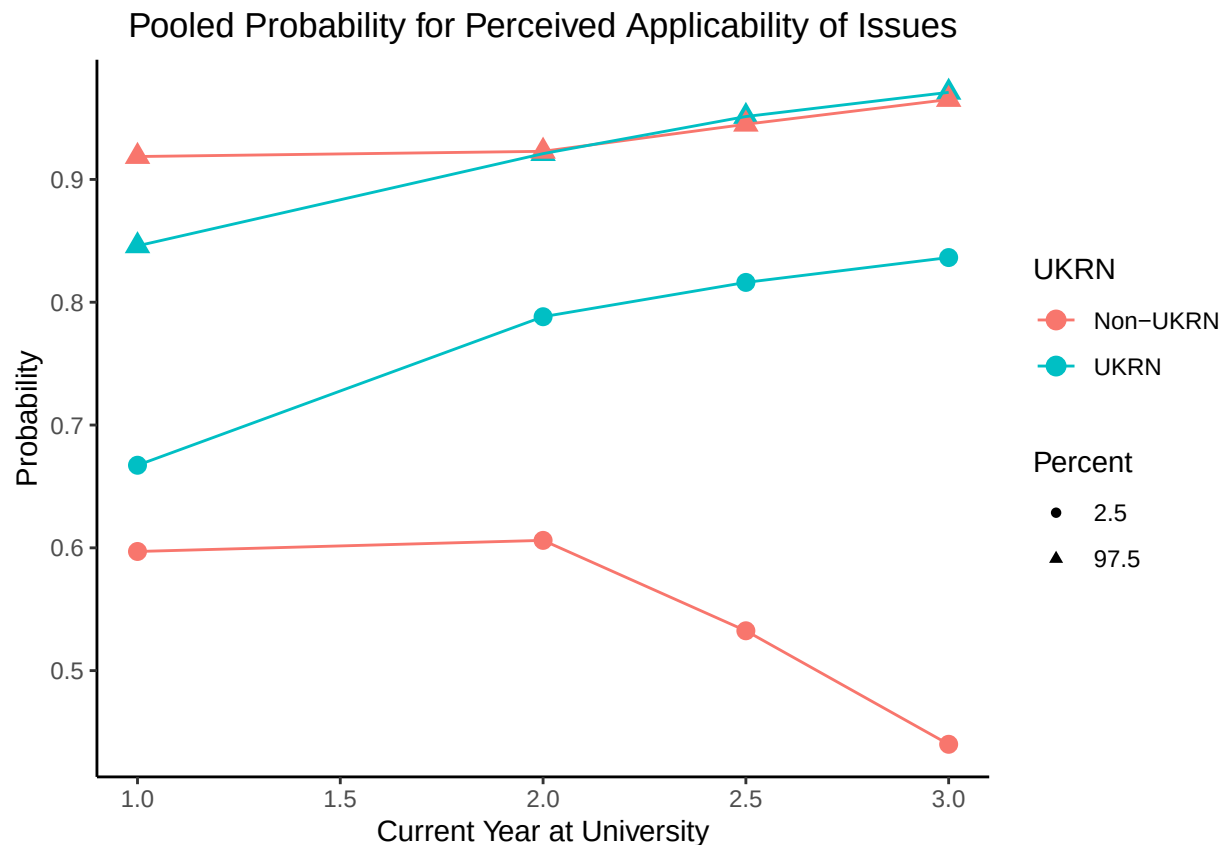
##          Rhat
## mu      1.00
## sigma   1.01
## alpha[1] 1.00
## alpha[2] 1.00
## alpha[3] 1.00
## alpha[4] 1.00
## alpha[5] 1.00
## alpha[6] 1.00
## alpha[7] 1.00
## beta1    1.00
## beta2    1.00
## beta3    1.00
## lp__     1.01
##
## Samples were drawn using NUTS(diag_e) at Thu Sep 26 09:26:35 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 525
 UKRN N = 473
 Non-UKRN N = 52
 Year 1 N = 316
 Year 2 N = 192
 Placement Year N = 0
 Year 4 N = 17

The output shows that all beta credibility intervals do cross 0. No significant results.



Data Set 2 - Truncated Poisson Regression

Awarweness Score Total

Bayesian regression for how much awareness of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement or third year

* Interaction between the two variables above

Inference for Stan model: anon_model.

4 chains, each with iter=10000; warmup=5000; thin=1;

post-warmup draws per chain=5000, total post-warmup draws=20000.

##

##	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff
## mu	2.37	0.01	0.27	1.85	2.19	2.37	2.54	2.90	2359
## sigma	0.19	0.00	0.10	0.07	0.12	0.17	0.23	0.46	4918
## alpha[1]	2.32	0.01	0.30	1.75	2.13	2.32	2.51	2.91	2418
## alpha[2]	2.21	0.01	0.29	1.63	2.02	2.21	2.39	2.79	2355
## alpha[3]	2.53	0.01	0.30	1.97	2.34	2.53	2.72	3.13	2366
## alpha[4]	2.41	0.01	0.29	1.84	2.22	2.41	2.60	3.00	2360
## alpha[5]	2.41	0.01	0.29	1.85	2.21	2.41	2.60	3.00	2767
## alpha[6]	2.38	0.00	0.20	2.01	2.25	2.38	2.51	2.77	2907
## alpha[7]	2.32	0.01	0.28	1.77	2.13	2.32	2.50	2.86	2813
## beta1	-0.66	0.01	0.31	-1.27	-0.85	-0.66	-0.46	-0.06	2421
## beta2	-0.21	0.00	0.13	-0.48	-0.30	-0.21	-0.12	0.05	2651

```

## beta3      0.61      0.00 0.15      0.32      0.51      0.61      0.71      0.91  3002
## lp__      3140.27      0.05 3.04 3133.30 3138.50 3140.68 3142.45 3145.09  4448
##          Rhat
## mu        1.01
## sigma     1.00
## alpha[1]  1.01
## alpha[2]  1.01
## alpha[3]  1.01
## alpha[4]  1.01
## alpha[5]  1.01
## alpha[6]  1.00
## alpha[7]  1.01
## beta1     1.01
## beta2     1.01
## beta3     1.00
## lp__      1.00
##
## Samples were drawn using NUTS(diag_e) at Thu Sep 26 10:13:59 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 525

UKRN N = 473

Non-UKRN N = 52

Year 1 N = 316

Year 2 N = 192

Placement Year N = 0

Year 4 N = 17

Var1	Freq
0	9
1	7
2	14
3	38
4	31
5	51
6	50
7	53
8	272

The output shows that beta credibility intervals do not cross 0 for Beta1.

2.5% = 0.3156397

97.5% = 0.9109164

The interpretation is that awareness of open and reproducible research is significantly more likely in students who are in non-UKRN universities.

The output shows that beta credibility intervals do not cross 0 for the interaction term.

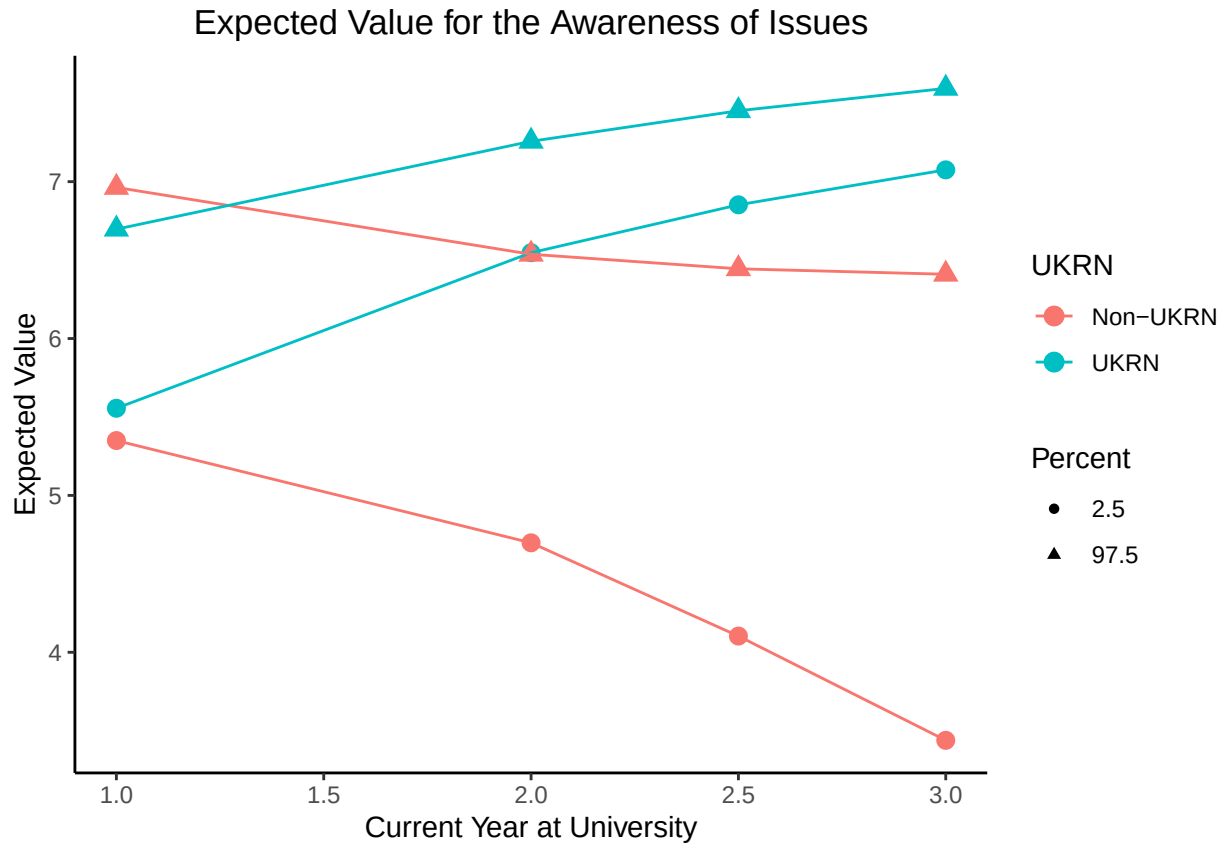
2.5% = 0.3156397

97.5% = 0.9109164

The interpretation is that students who attend a UKRN affiliated university are more likely to be aware of

the issues around open and reproducible research in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.

There were 4 divergent transitions after warmup.



Experience Score Total

Bayesian regression for how much experience of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement or third year

* Interaction between the two variables above

Inference for Stan model: anon_model.

4 chains, each with iter=10000; warmup=5000; thin=1;

post-warmup draws per chain=5000, total post-warmup draws=20000.

##

##	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff
## mu	0.38	0.01	0.38	-0.42	0.16	0.41	0.64	1.06	2721
## sigma	0.29	0.00	0.16	0.10	0.19	0.26	0.35	0.70	4070
## alpha[1]	0.19	0.01	0.43	-0.72	-0.06	0.22	0.47	0.93	2628
## alpha[2]	0.43	0.01	0.41	-0.45	0.19	0.46	0.70	1.17	2602
## alpha[3]	0.61	0.01	0.41	-0.27	0.37	0.63	0.88	1.34	2612
## alpha[4]	0.30	0.01	0.42	-0.59	0.06	0.33	0.58	1.04	2595
## alpha[5]	0.32	0.01	0.41	-0.56	0.06	0.34	0.60	1.05	3112
## alpha[6]	0.63	0.00	0.23	0.17	0.47	0.63	0.78	1.08	3535

```

## alpha[7]    0.20    0.01 0.42   -0.71   -0.07    0.23    0.49    0.95  3034
## beta1       0.14    0.01 0.42   -0.61   -0.14    0.12    0.39    1.03  2661
## beta2       0.11    0.00 0.19   -0.25   -0.01    0.11    0.24    0.49  3001
## beta3       0.10    0.00 0.19   -0.28   -0.03    0.10    0.23    0.48  3176
## lp__        -158.72  0.05 2.91 -165.35 -160.45 -158.37 -156.63 -154.04 4115
##              Rhat
## mu           1
## sigma        1
## alpha[1]     1
## alpha[2]     1
## alpha[3]     1
## alpha[4]     1
## alpha[5]     1
## alpha[6]     1
## alpha[7]     1
## beta1        1
## beta2        1
## beta3        1
## lp__         1
##
## Samples were drawn using NUTS(diag_e) at Thu Sep 26 10:42:37 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

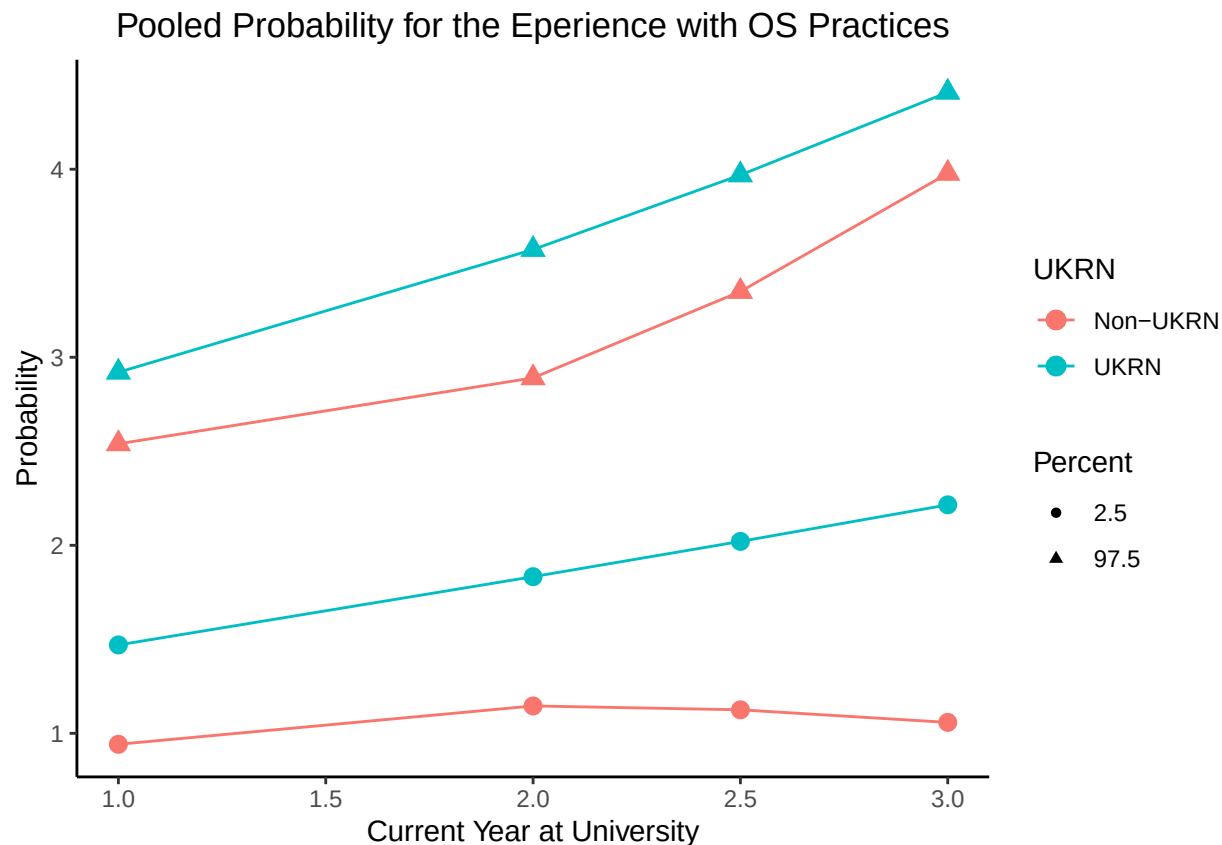
Number of participants in the analysis

Total N = 525
 UKRN N = 473
 Non-UKRN N = 52
 Year 1 N = 316
 Year 2 N = 192
 Placement Year N = 0
 Year 4 N = 17

Var1	Freq
0	113
1	81
2	101
3	84
4	73
5	45
6	19
7	7
8	2

The output shows that all beta credibility intervals cross 0.

There were 4 divergent transitions after warmup.



Data Set 2 - Linear Regression

Conceptual Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4.

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd  2.5%  25%   50%   75%  97.5% n_eff Rhat
## mu           0.47    0.00 0.17  0.14  0.36  0.47  0.58  0.79  2597   1
## sigma        0.41    0.00 0.01  0.39  0.40  0.41  0.42  0.44 10131   1
## sigma_a      0.11    0.00 0.07  0.04  0.07  0.10  0.13  0.27  4746   1
## alpha[1]     0.46    0.00 0.18  0.09  0.34  0.46  0.58  0.82  2603   1
## alpha[2]     0.55    0.00 0.18  0.20  0.44  0.55  0.66  0.91  2543   1
## alpha[3]     0.49    0.00 0.18  0.13  0.38  0.49  0.61  0.85  2519   1
## alpha[4]     0.37    0.00 0.18  0.00  0.26  0.37  0.48  0.72  2541   1
## alpha[5]     0.50    0.00 0.18  0.14  0.38  0.49  0.62  0.87  3017   1
## alpha[6]     0.47    0.00 0.13  0.22  0.38  0.47  0.55  0.72  3308   1
## alpha[7]     0.44    0.00 0.18  0.08  0.32  0.44  0.56  0.78  3095   1
## beta1        0.18    0.00 0.19 -0.19  0.06  0.18  0.30  0.55  2612   1
```

```
## beta2      0.07      0.00 0.09 -0.11  0.01  0.07  0.13  0.25 3041  1
## beta3     -0.07      0.00 0.10 -0.26 -0.13 -0.07  0.00  0.12 3278  1
## lp__      216.26     0.05 3.15 209.15 214.38 216.64 218.54 221.33 4022  1
##
## Samples were drawn using NUTS(diag_e) at Thu Sep 26 11:38:22 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 525

UKRN N = 473

Non-UKRN N = 52

Year 1 N = 316

Year 2 N = 192

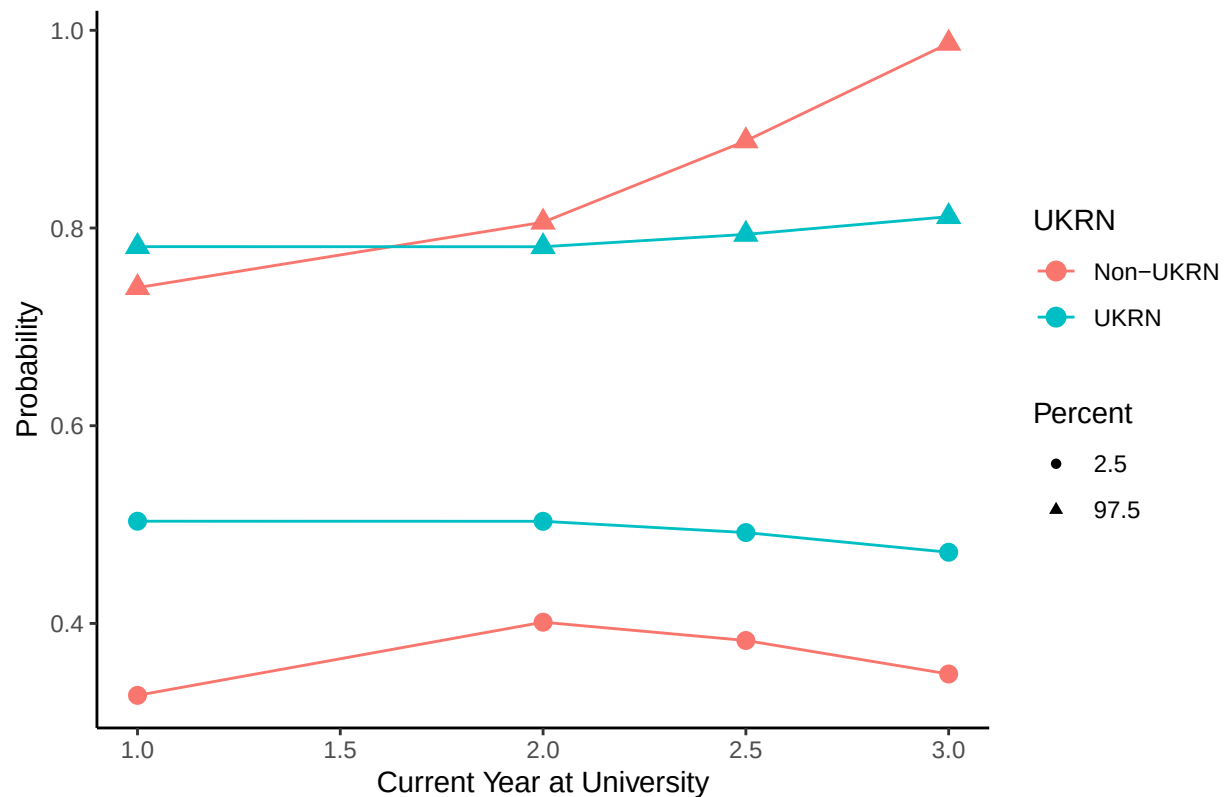
Placement Year N = 0

Year 4 N = 17

The output shows that all beta credibility intervals cross 0.

6 divergent transitions.

Expected Value for positive Conceptual Perceptions of OS Practices



Situational Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4.

The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN
 * Whether the participant was in their first, second, placement or third year
 * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd  2.5%   25%   50%   75%  97.5% n_eff Rhat
## mu           0.12    0.00  0.22 -0.33  -0.02   0.13   0.27   0.54  2827   1
## sigma        0.51    0.00  0.02  0.48   0.50   0.51   0.52   0.54 10506   1
## sigma_a      0.15    0.00  0.09  0.05   0.09   0.13   0.18   0.39  4402   1
## alpha[1]     0.09    0.00  0.24 -0.41  -0.06   0.10   0.25   0.55  2838   1
## alpha[2]     0.22    0.00  0.24 -0.27   0.07   0.23   0.38   0.67  2807   1
## alpha[3]     0.16    0.00  0.24 -0.33   0.01   0.17   0.32   0.61  2812   1
## alpha[4]     0.02    0.00  0.24 -0.50  -0.14   0.02   0.18   0.46  2769   1
## alpha[5]     0.16    0.00  0.23 -0.31   0.00   0.16   0.31   0.60  3400   1
## alpha[6]     0.20    0.00  0.16 -0.11   0.09   0.20   0.30   0.51  3576   1
## alpha[7]     0.02    0.00  0.24 -0.48  -0.13   0.03   0.19   0.48  3227   1
## beta1        0.16    0.00  0.25 -0.30   0.00   0.16   0.32   0.68  2843   1
## beta2        0.05    0.00  0.12 -0.18  -0.03   0.05   0.12   0.27  3186   1
## beta3        0.04    0.00  0.12 -0.20  -0.04   0.04   0.12   0.28  3430   1
## lp__        105.20    0.05  3.23  97.86 103.27 105.57 107.52 110.44 4194   1
##
## Samples were drawn using NUTS(diag_e) at Thu Sep 26 11:53:02 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 525
 UKRN N = 473
 Non-UKRN N = 52
 Year 1 N = 316
 Year 2 N = 192
 Placement Year N = 0
 Year 4 N = 17

The output shows that all beta credibility intervals cross 0.

11 divergent transitions.

Expected Value for positive Conceptual Perceptions of OS Practices

